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Neural Networks and Deep Learning

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2026-11-09

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"Biology / Biomedicine is a Data Science."

— W. Evan Johnson

Addendum: don't avoid the math!

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"With great power comes great responsibility."

— Uncle Ben, *Spider-Man*

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Learning objectives

By the end of this lecture you should be able to:

- Describe an artificial neuron: weights, bias, activation function
- Explain how a network learns: loss, backpropagation, gradient descent
- Fit and tune a multilayer perceptron in R (`neuralnet`, `nnet` via `caret`)
- Explain what convolutional and recurrent networks add, and why
- Describe the key idea of transformers: attention

A quick map of “AI”

Where do neural networks fit in the broader AI landscape?

- **Narrow AI** (all AI today): does one specific task well — Face ID, spam filters, recommendations, Google Maps, ChatGPT
- **Generative AI**: creates new content from learned patterns — text, images, music, video, voice
- **Agentic AI** (emerging): takes actions, makes plans, and uses tools to accomplish goals — schedules meetings, sends email, runs tasks automatically

A quick map of "AI"

- **Artificial General Intelligence (AGI)**: human-level reasoning across many tasks — **theoretical; does not exist** (movies exaggerate it)
- **Artificial Super Intelligence (ASI)**: hypothetical intelligence that would *surpass* humans across virtually all domains — **also does not exist**

Everything we build today — including the neural networks in this lecture — is **Narrow AI**.

Real (human) intelligence

The ability to think, choose, create, understand meaning, and act with moral responsibility:

- **Agency** — we choose
- **Moral judgment** — right vs. wrong
- **Creativity** — truly new ideas
- **Understanding meaning**
- **Empathy and relationships**
- **Spiritual capacity**

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"AI will not replace us, it will amplify us."

— Pedro Domingos

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From neurons to networks

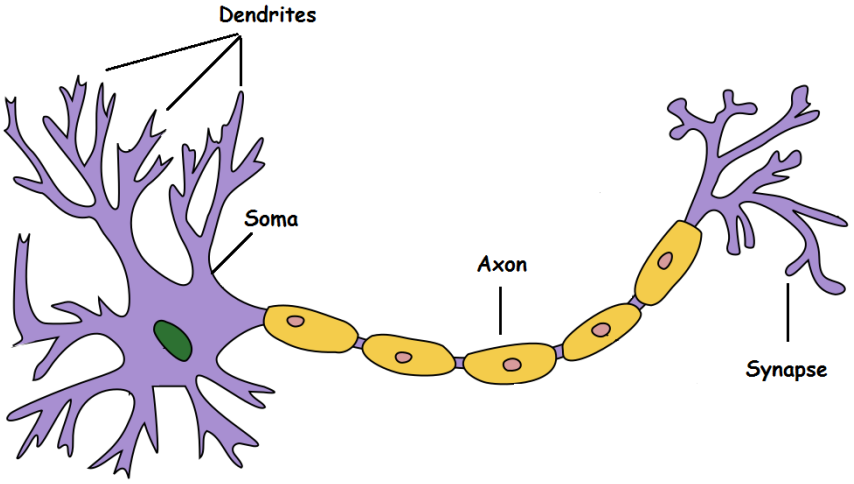
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Biological neurons

The brain has billions of neurons, each a simple processor, wired into a **biological neural network** that passes electrical signals from neuron to neuron:

- **Dendrites** — receive input signals
- **Soma** (cell body) — sums the inputs
- **Axon** $\rightarrow$ **synapses** — pass the output on to other neurons

Biological neurons



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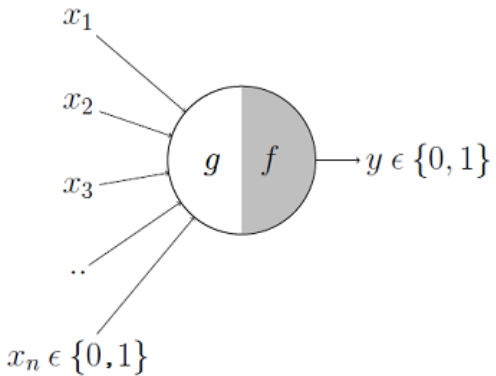
McCulloch & Pitts (1943)

The first mathematical model of a neuron:

- Binary inputs, fixed weights, and a **threshold**: fire (1) or not (0)
- **No layers and no training rule**
- Networks of such units can compute logical functions: the brain as a computing device

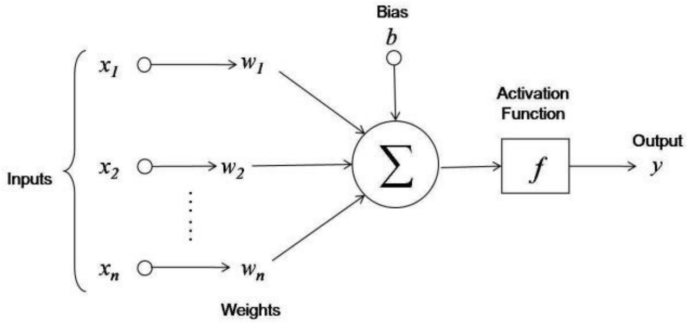
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McCulloch & Pitts (1943)



The perceptron (Rosenblatt, 1957–58)

The first **trainable** neural network: one layer of adjustable weights and a threshold, learned from labeled examples.



A short history

- **1943:** McCulloch & Pitts neuron
- **1958:** Rosenblatt's perceptron
- **1969:** Minsky & Papert, *Perceptrons*: a single-layer perceptron can't learn simple functions like XOR $\rightarrow$ interest collapses
- **1986:** Rumelhart, Hinton & Williams popularize **backpropagation** for multilayer networks
- **2012:** AlexNet (deep CNN + GPUs) wins ImageNet $\rightarrow$ the **deep learning** era

An artificial neuron

$$\text{output} = g\Big(\sum_{j=1}^{p} w_j x_j + b\Big)$$

- **Inputs** $x_1, \ldots, x_p$: the predictors
- **Weights** w_j and **bias** b: the parameters we learn
- **Activation function** g: maps the weighted sum to the output

With g = logistic, a single neuron **is logistic regression**.

Activation functions

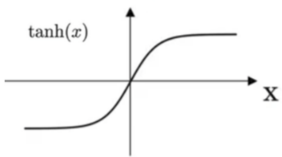
The activation function gives the network its **non-linearity**:

- **Identity**: output = input (linear; regression outputs)
- **Binary step**: on/off above a threshold (the original perceptron)
- **Sigmoid / tanh**: smooth S-curves; sigmoid gives probabilities
- **ReLU** $\max(0, x)$: the modern default for hidden layers
- **Softmax**: turns K outputs into class probabilities

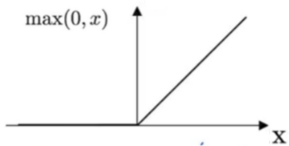
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Activation functions

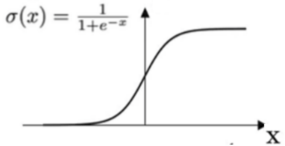
Hyper Tangent Function



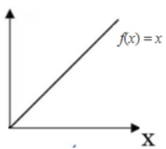
ReLU Function



Sigmoid Function

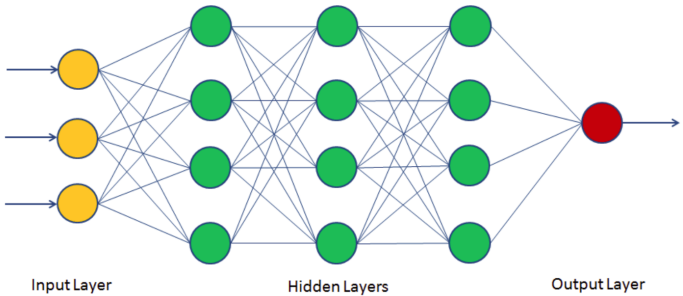


Identity Function



Multilayer networks

A **multilayer perceptron** (MLP) stacks layers of neurons: input $\rightarrow$ hidden layer(s) $\rightarrow$ output. "**Deep**" = many hidden layers.



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How networks learn

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How does a neural network learn?

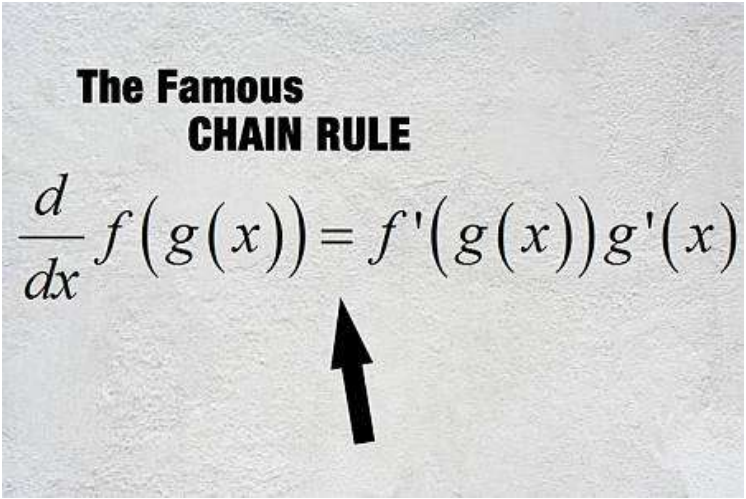
Training means finding the **weights** that make predictions match the data:

1. A **loss** $\ell(y, \hat{y})$: how wrong is the prediction? (squared error, cross-entropy)
2. **Backpropagation**: how does each weight affect the loss?
3. **Gradient descent**: nudge each weight to reduce the loss

"AI is not magic; it's math."

— Peter Norvig

The chain rule



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The chain rule

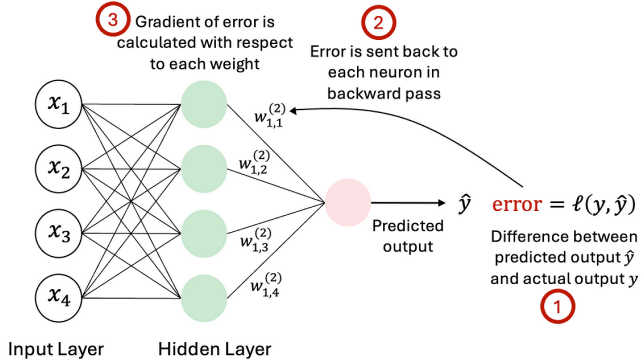
A neural network is **functions composed inside functions**. The derivative of a composition multiplies the derivatives of each layer:

$$f' \circ g \circ h \circ k(x) \times g' \circ h \circ k(x) \times h' \circ k(x) \times k'(x)$$



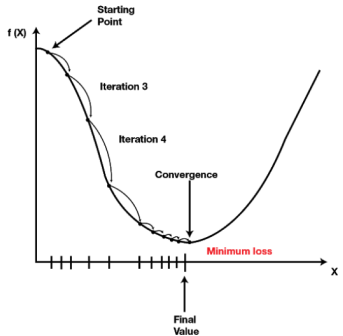
Backpropagation

The chain rule sends the error **backward** through the network, giving the gradient of the loss for **every weight**:



Gradient descent

Update each weight against its gradient: $w \leftarrow w - \eta \, \partial \ell / \partial w$ (learning rate η). In practice, **stochastic** gradient descent uses small **mini-batches**; one pass through the data is an **epoch**.



Training in practice

Networks have many weights, so they **overfit** easily:

- **Weight decay** (L2 penalty): regularization again (`decay` in `nnet`)
- **Early stopping**: stop when validation loss stops improving
- **Dropout**: randomly switch off neurons during training
- **Scale the inputs**; start from **random weights** (results vary by seed)

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Neural networks in R

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A toy example: job placement

Technical knowledge (TKS) and communication skill (CSS) scores (rescaled to 0–1), and placement:

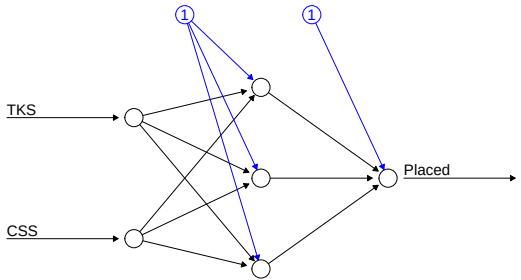
```
df <- data.frame(TKS    = c(20, 10, 30, 20, 80, 30) / 100,
                 CSS    = c(90, 20, 40, 50, 50, 80) / 100,
                 Placed = c(1, 0, 0, 0, 1, 1))
df
```

```
##   TKS CSS Placed
## 1 0.2 0.9      1
## 2 0.1 0.2      0
## 3 0.3 0.4      0
## 4 0.2 0.5      0
## 5 0.8 0.5      1
## 6 0.3 0.8      1
```

A toy example: job placement

A multilayer perceptron with one hidden layer of 3 neurons:

```
set.seed(0)
nn <- neuralnet(Placed ~ TKS + CSS, data = df, hidden = 3,
                act.fct = "logistic", linear.output = FALSE)
```



A toy example: job placement

Predict for three new students; threshold at 0.5:

```
test <- data.frame(TKS = c(30, 40, 85) / 100, CSS = c(85, 50, 40) / 100)
prob <- predict(nn, test)
cbind(test, prob = round(prob, 3), pred = ifelse(prob > 0.5, 1, 0))
```

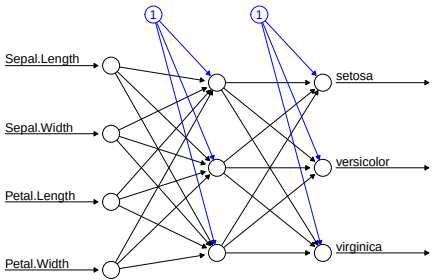
```
##    TKS  CSS  prob pred
## 1 0.30 0.85 0.942    1
## 2 0.40 0.50 0.442    0
## 3 0.85 0.40 0.903    1
```

With 6 training points, this is an illustration, not a model!

Iris: three classes

`neuralnet` fits one output neuron per class (inputs scaled first; unscaled, it fails to converge):

```
iris_s <- mutate(iris, across(where(is.numeric), ~ as.numeric(scale(.x))))
set.seed(0)
nn_iris <- neuralnet(Species ~ ., data = iris_s, hidden = 3, linear.output = FALSE
plot(nn_iris, rep = "best", information = FALSE, show.weights = FALSE)
```



Tuning with caret: `nnet`

`method = "nnet"`: one hidden layer; tune its **size** and the weight **decay**.
Train/test split, as always:

```
set.seed(0)
idx <- createDataPartition(iris$Species, p = 0.7, list = FALSE)
nn_caret <- train(Species ~ ., data = iris[idx, ], method = "nnet",
                  preProcess = c("center", "scale"), trace = FALSE)
nn_caret$bestTune
```

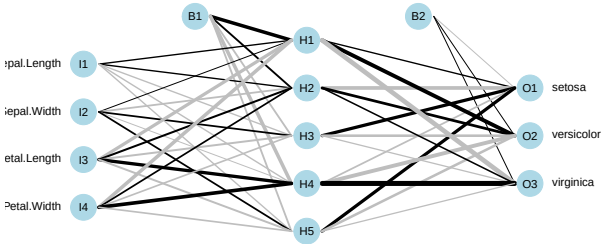
```
##   size decay
## 9    5   0.1
```

```
confusionMatrix(predict(nn_caret, iris[-idx, ]),
                iris$Species[-idx])$overall["Accuracy"]
```

```
##  Accuracy
## 0.9555556
```

Tuning with caret: nnet

```
NeuralNetTools::plotnet(nn_caret$finalModel)
```



Biomarker development revisited: TB

The TB NanoString data from the SVM lecture, same split and CV:

```
tb <- readRDS("TBnanostring.rds")
tb$TB_Status <- factor(tb$TB_Status, levels = c("LTBI", "TB"))
set.seed(1)
idx <- createDataPartition(tb$TB_Status, p = 0.7, list = FALSE)
tb_train <- tb[idx, ]; tb_test <- tb[-idx, ]
ctrl <- trainControl(method = "cv", number = 10, classProbs = TRUE,
                     summaryFunction = twoClassSummary)
set.seed(1)
nn_tb <- train(TB_Status ~ ., data = tb_train, method = "nnet",
               preProcess = c("center", "scale"), metric = "ROC", trControl = ctrl,
               tuneGrid = expand.grid(size = c(1, 3, 5), decay = c(0.01, 0.1, 1, 2)),
               trace = FALSE, maxit = 500)
nn_tb$bestTune
```

```
##   size decay
## 8    3     2
```

Biomarker development revisited: TB

```
p_tb <- predict(nn_tb, tb_test, type = "prob")[, "TB"]
auc(roc(tb_test$TB_Status, p_tb, levels = c("LTBI", "TB"),
        direction = "<", quiet = TRUE))
```

```
## Area under the curve: 0.9319
```

```
confusionMatrix(predict(nn_tb, tb_test), tb_test$TB_Status,
                positive = "TB")$byClass[c("Sensitivity", "Specificity")]
```

```
## Sensitivity Specificity
##   0.8695652   0.9333333
```

Test AUC $\approx$ the SVMs (0.92–0.93), with **strong weight decay** chosen. With 125 samples and 107 genes, a neural net has no edge.

When does deep learning shine?

- **Large** data sets (thousands to millions of examples)
- **Raw, structured inputs**: images, sequences, text, audio
- The network **learns the features** that we would otherwise hand-engineer
- For small **tabular** data (most clinical and omics studies), regularized regression, forests, and boosting are usually as good and easier to tune

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Network architectures

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Feedforward vs. recurrent

- **Feedforward**: signals travel one way, layer to layer, input to output (the MLP)
- **Recurrent**: contains a loop: the hidden state from the previous step is fed back in, giving the network **memory**
- **Convolutional**: shares weights across positions in an image or sequence

Others: LSTM/GRU, sequence-to-sequence, radial basis function networks, autoencoders, transformers.

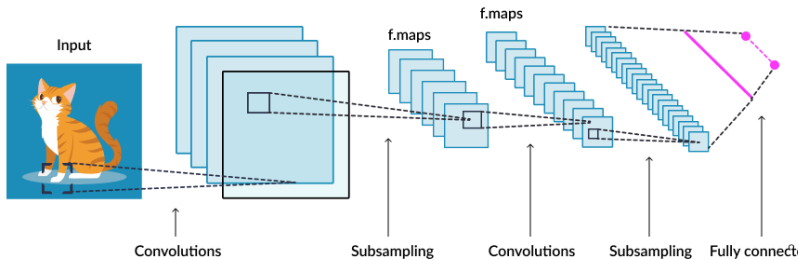
Convolutional neural networks

A **CNN** is specialized for **grid-like data** such as images:

- Small **filters (kernels)** with **shared weights** slide across the input $\rightarrow$ **feature maps**
- A pattern is detected **wherever it appears** (translation equivariance; pooling adds approximate invariance)
- Far fewer weights than a fully connected MLP: a built-in **regularization**

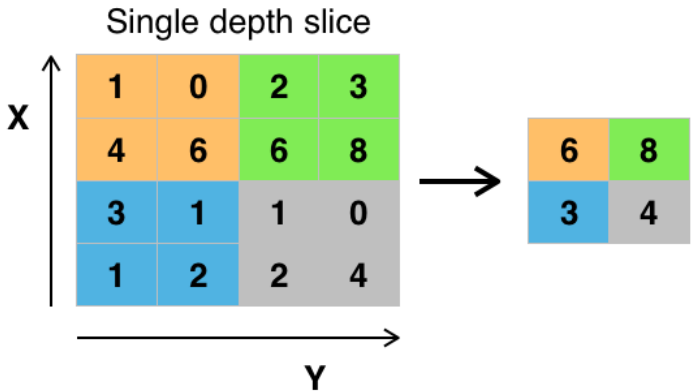
Convolutional neural networks

- **Filters are learned** from data, not hand-engineered
- Early layers find edges; later layers find shapes and objects
- Uses: medical imaging, pathology slides, video, genomic sequences



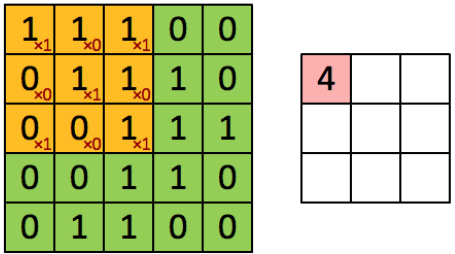
Convolutional neural networks

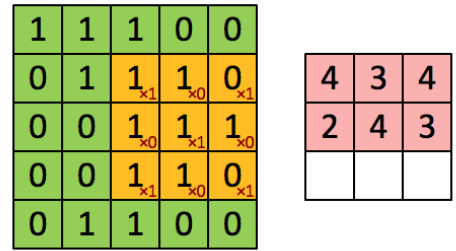
Pooling (here, max pooling) shrinks each feature map:



Convolutional neural networks

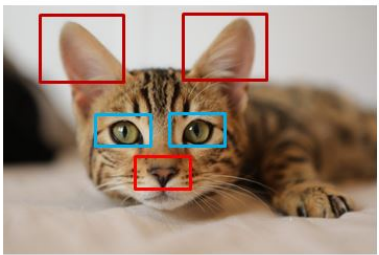
A sliding (overlapping) filter: **kernel**, **stride**, and **padding**:

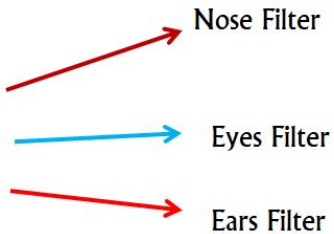




Convolutional neural networks

Now applying this to cats:





Convolutional neural networks

More realistically:



Cat



Cat



Representation

Convolutional neural networks

And as a classifier:





Recurrent neural networks

- For **sequential data**: text, time series, speech, sequences
- Standard networks assume **independent inputs**, but sequences have **dependencies**: in *"The cat sat on the ____"*, you need the previous words
- A **hidden state** carries memory from one step to the next

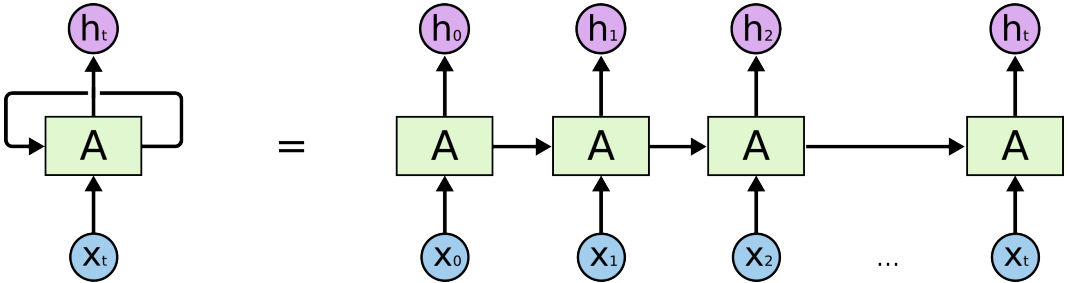
RNN architecture

The same weights are applied at every time step t:

- Input: x_t
- Hidden state: $h_t = f(Wx_t + Uh_{t-1} + b)$
- Output: $y_t = g(Vh_t + c)$

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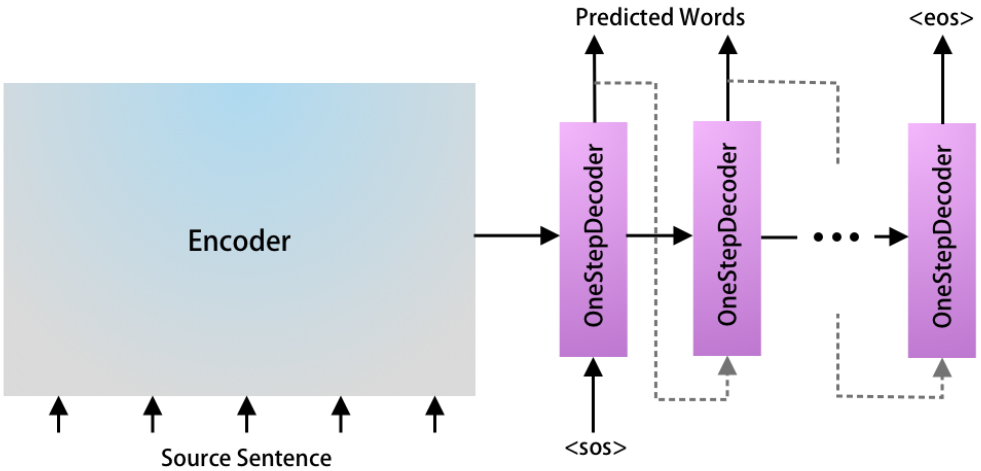
Diagram of an RNN (unrolled)



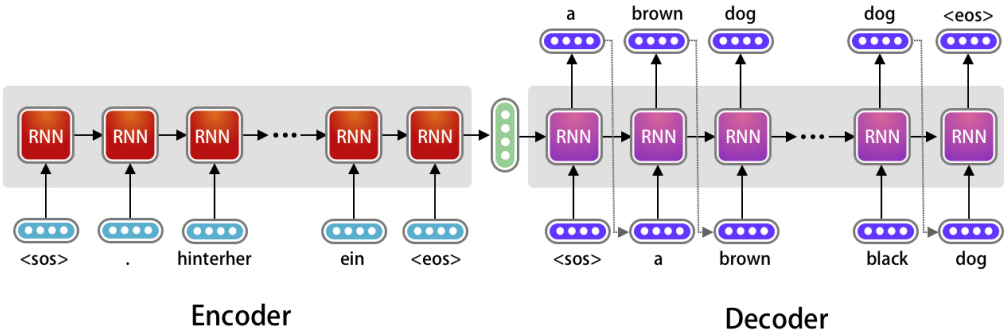
Challenges of RNNs

- **Vanishing/exploding gradients**: backpropagating through many time steps multiplies many derivatives
- Hard to learn long-range dependencies
- Fixes: **LSTM** (long short-term memory) and **GRU** (gated recurrent unit) cells

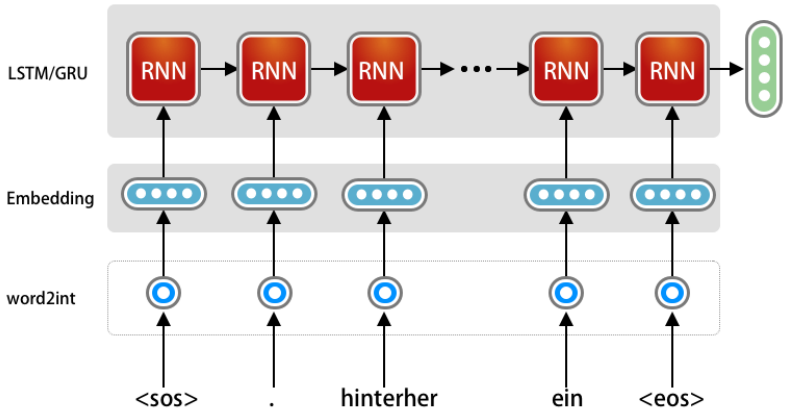
RNN for language translation



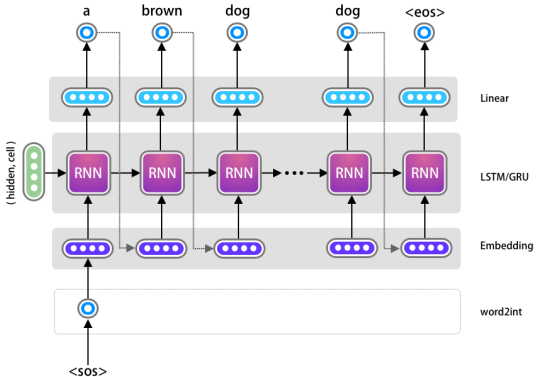
RNN for language translation



RNN for language translation



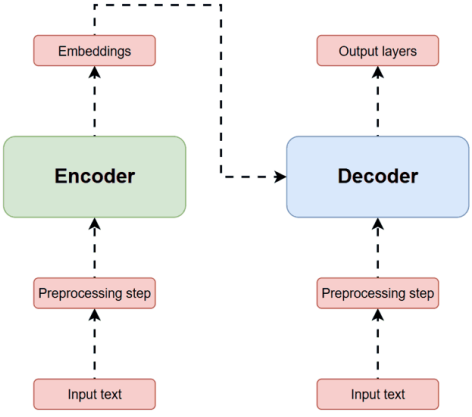
RNN for language translation



From RNNs to transformers

- RNNs process a sequence **one step at a time**: slow, and memory fades
- **Transformers** process the **entire sequence at once**, using **attention**
- Introduced in *"Attention Is All You Need"* (Vaswani et al., 2017)

Transformers: encoder–decoder



Key idea: attention

- Each token **looks at every other token** and decides which ones matter
- Example: in translation, each output word attends to the relevant source words
- For queries Q, keys K, values V:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

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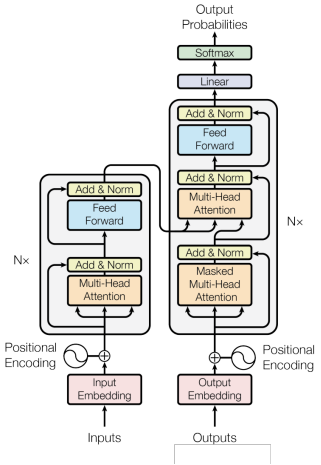
Transformer architecture

- **Multi-head** self-attention: several attention "heads" learn different relationships
- **Positional encoding** adds word order (there is no recurrence)
- Each block: attention + feedforward network, with residual connections and layer normalization
- Original transformer: **encoder + decoder** (translation). **BERT**: encoder only. **GPT**: decoder only.

Transformer architecture

BERT

Encoder



GPT

Decoder

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GPT: a very advanced autocomplete

GPT = **G**enerative **P**retrained **T**ransformer. At its core, it predicts the next word (token):

- Looks at **all** the previous words at once
- Assigns **probabilities** to possible next words
- Picks a likely next word, then repeats

More on generative AI and LLMs in the coming weeks.

Summary

- A neuron: weighted sum + bias + non-linear activation; an MLP stacks layers of them
- Training: a loss, backpropagation (chain rule), and gradient descent
- Networks overfit easily: weight decay, early stopping, dropout; tune by CV
- On small tabular data (TB NanoString), an MLP matched but didn't beat SVMs
- CNNs share filters across images; RNNs carry memory along sequences
- Transformers replace recurrence with attention: the basis of today's LLMs

Python companion

Python versions of these examples (`scikit-learn`'s `MLPClassifier`):

`python/week7/neuralNet.ipynb`

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Center for Data Science

Session Info

```
sessionInfo()
```

```
## R version 4.3.3 (2024-02-29)
## Platform: x86_64-pc-linux-gnu (64-bit)
## Running under: Ubuntu 24.04.5 LTS
##
## Matrix products: default
## BLAS:   /usr/lib/x86_64-linux-gnu/blas/libblas.so.3.12.0
## LAPACK: /usr/lib/x86_64-linux-gnu/lapack/liblapack.so.3.12.0
##
## locale:
## [1] C
##
## time zone: America/Denver
## tzcode source: system (glibc)
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] pROC_1.18.5      nnet_7.3-19      neuralnet_1.44.2 lubridate_1.9.3
##  [5] forcats_1.0.0    stringr_1.5.1    dplyr_1.1.4      purrr_1.0.2
##  [9] readr_2.1.5      tidyr_1.3.1      tibble_3.2.1     tidyverse_2.0.0
## [13] caret_6.0-94     lattice_0.22-5   ggplot2_3.4.4
##
## loaded via a namespace (and not attached):
##  [1] gtable_0.3.4         xfun_0.41            recipes_1.0.9
##  [4] tzdb_0.4.0           vctrs_0.6.5          tools_4.3.3
```